Group II, Row IV with $d_\perp$ kept general ($\delta^{mm} = d_\perp$)

1 Result

Assumptions.

- $d_\perp$ is never set to 2. Every transverse trace is $\delta^{mm} = \delta^{ii} = d_\perp$, both in the polarization sums and in the tensor z integral.
- The z integral, the only divergent one, is done in $d = 2 - \epsilon$ dimensions, in $\overline{\text{MS}}$, in the scheme used for Group I, Row II and Row V, part 2b.
- No limit is taken in $\vee$, k^+ or Λ.

Notation. $P \equiv p^+ + k^+$ and $\xi \equiv k^+/P$, the fraction of the parent carried by the measured gluon. Further,

$$\Delta \equiv y' - y, \qquad \hat{\Delta} \equiv \frac{\Delta}{|\Delta|}, \qquad X \equiv \mathcal{K}(x-y) = \frac{x-y}{(x-y)^2}, \qquad X' \equiv \mathcal{K}(x'-y'), \tag{1.1}$$

$$\mathcal{O} \equiv \big[U^{ce'}(y') - U^{ce'}(x')\big]\rho^{e'}(x')\,\big[U^{ce}(y) - U^{ce}(x)\big]\rho^{e}(x) \tag{1.2}$$

(with $U^{bc} = U^{cb}$ this is your $[U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$), and

$$E \equiv \frac{2}{\epsilon} + 2\gamma_E + \log\frac{\Delta^2\mu^2_{\overline{MS}}}{4} = -\Big[\frac{1}{\bar\epsilon} - \log\frac{\Delta^2\mu^2_{\overline{MS}}e^{2\gamma_E}}{4}\Big] \equiv -\bar{E}. \tag{1.3}$$

The pole is infrared ($\epsilon < 0$); $\frac{1}{\bar\epsilon} = -\frac{2}{\epsilon}$ is your convention.

Result. The splitting function is

$$\hat\Phi(\xi) \equiv \xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}, \tag{1.4}$$

and Row IV is

$$\boxed{\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot\Delta/\xi} \\ \times\Big\{X\cdot X'\Big[d_\perp\,\hat\Phi(\xi)\,\bar E - 2\Big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\Big] - 2\,d_\perp\,\xi(1-\xi)\,(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big\}\,\mathcal{O}.\end{aligned}} \tag{1.5}$$

- $d_\perp$ enters in two places: it multiplies the whole pole term $\hat\Phi\bar E$, and the $(X\cdot\hat\Delta)(X'\cdot\hat\Delta)$ term.
- The term $-2(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi})$ does not depend on $d_\perp$. It comes from the traceless part of the z integral, contracted with $\delta_{mm'}$ (Sec. 2.6).
- The term $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km})$ of your bracket does not contribute, for any $d_\perp$.

In E form, Eq. (1.5) is $-\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\int\frac{d\xi}{\xi^2}\int e^{-ik\cdot\Delta/\xi}\{X\cdot X'[d_\perp\hat\Phi E + 2(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi})] + 2d_\perp\xi(1-\xi)(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\}\mathcal{O}$.

Soft endpoint ($\xi \to 1$). Write $\xi_0 = \frac{k^+}{\vee}$, $\xi_1 = \frac{k^+}{k^++\Lambda}$ and $g(\xi) = \xi^{-2}e^{-ik\cdot\Delta/\xi}$. Exactly,

$$\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log\Big(1-\frac{k^+}{\vee}\Big) + \log\Big(1+\frac{\Lambda}{k^+}\Big)\Big]\int_{x,x',y,y'} e^{-ik\cdot\Delta}\,X\cdot X'\,\big(d_\perp\bar E - 2\big)\,\mathcal{O} \\ &+ \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\int_{\xi_0}^{\xi_1}d\xi\int_{x,x',y,y'}\Big\{X\cdot X'\Big[\big(d_\perp\bar E - 2\big)\frac{g(\xi)-g(1)}{1-\xi} \\ &+ g(\xi)\Big(d_\perp\bar E\Big(\xi(1-\xi) - 1 + \frac{1-\xi}{\xi}\Big) - 2\Big(\frac{1-\xi}{\xi} - 1\Big)\Big)\Big] - 2d_\perp\xi(1-\xi)\,g(\xi)\,(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big\}\mathcal{O}.\end{aligned} \tag{1.6}$$

The coefficient of the endpoint logarithm is $d_\perp\bar E - 2$.

Pole and LO. The LO cross section (emitter at y) is

$$\frac{d^3N_{\rm LO}}{d^3k}(k^+,\mathbf{k}) = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,x',y,y'} e^{-ik\cdot\Delta} X\cdot X'\,\mathcal{O}. \tag{1.7}$$

With it, the pole of Eq. (1.5) is

$$\left.\frac{d^3N_{\rm IV}}{d^3k}\right|_{\rm pole} = \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^3}\frac{d_\perp}{2}P_{gg}^{(r)}(\xi)\,\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big), \qquad P_{gg}^{(r)}(\xi) = 2N_c\hat\Phi(\xi). \tag{1.8}$$

So the residue is $\frac{d_\perp}{2}P_{gg}^{(r)}$: the polarization trace multiplies the real part of the splitting function.

The two usual choices.

- $d_\perp = 2-\epsilon$ (conventional dimensional regularization, as in Group I Rows II and V). Then $d_\perp\bar E = 2\bar E + 2$, and the braces of (1.5) become $2\{X\cdot X'[\hat\Phi\bar E + \xi(1-\xi)] - 2\xi(1-\xi)(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\}$. This is the result I gave before, in the previous version of `Group_II_corrected.pdf`, where $d_\perp = 2-\epsilon$ had been inserted.
  - Up to the corrections given earlier (the constant in the logarithm, the ξ limits), it also agrees with your Eq. (1.15): there the "+1" in $[\frac{1}{\bar\epsilon}+1-\log]$ is $d_\perp\bar E$ with $d_\perp = 2-\epsilon$, while the last line keeps $d_\perp$. Note that $\int_1^{k^+/\vee} = -\int_{k^+/\vee}^1$.
  - The endpoint coefficient becomes $2\bar E$.
- $d_\perp = 2$ (polarizations in two dimensions). The braces become $2\{X\cdot X'[\hat\Phi\bar E - \frac{\xi}{1-\xi} - \frac{1-\xi}{\xi}] - 2\xi(1-\xi)(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\}$ and the endpoint coefficient is $2\bar E - 2$.

The difference between the two choices is finite, $2\hat\Phi\, X\cdot X'$ in the braces. The same choice of $d_\perp$ must be used in every row, so that the poles and the finite parts match between rows and with the counterterm.

2 Derivation

2.1 Step 1: the row

Row IV is the term $4\,\mathfrak{C}'^\dagger\mathfrak{C}'$ of Eq. (CS2). The parent gluon P is emitted by the charge at x (x' in the conjugate amplitude). It crosses the target at y (y') and splits after the target into the measured gluon k^+ at w (w') and the unobserved gluon p^+ at z. Inserting

- $\mathbb{A}^{(1)}(P,-y) = \frac{ig}{\sqrt{2\pi}\sqrt{P}}\int_x \mathcal{K}(x-y)\rho(x)$,
- and $\mathbb{C}_1$, with vertex bracket

$$V_{ij;km} \equiv \delta_{km}\delta_{ij} - \frac{P}{p^+}\delta_{jm}\delta_{ik} - \frac{P}{k^+}\delta_{im}\delta_{kj} \tag{2.1}$$

  (i: measured gluon, j: unobserved gluon, k: parent, m: index of the splitting kernel),

gives your Eq. (1.12):

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik\cdot(w'-w)}\int_{x,x',y,y'}\mathcal{K}^m(w-z)\mathcal{K}^k(x-y)\mathcal{K}^{m'}(w'-z)\mathcal{K}^{k'}(x'-y')\frac{p^+k^+}{P^4}$$
$$\times\,\delta^{(d)}\Big[y-w+\frac{p^+}{P}(w-z)\Big]\delta^{(d)}\Big[y'-w'+\frac{p^+}{P}(w'-z)\Big]\,V_{ij;km}V_{ij;k'm'}\times[\text{colour}]. \tag{2.2}$$

The prefactor is $4\times|\mathbb{A}^{(1)}(P)|^2\times|\mathbb{C}_1|^2$: $4\cdot\frac{g^2}{2\pi^2 P}\cdot\frac{g^2}{8\pi^2 P}\frac{p^+k^+}{P^2} = \frac{g^4}{4\pi^4}\frac{p^+k^+}{P^4}$.

2.2 Step 2: the polarization sum ($\delta^{ii} = d_\perp$).

The measured and the unobserved gluon are summed over i and j. The nine products give:

product	$\sum_{ij}$	result
$\delta_{km}\delta_{ij}\cdot\delta_{k'm'}\delta_{ij}$	$\delta_{ij}\delta_{ij} = d_\perp$	$d_\perp\,\delta_{km}\delta_{k'm'}$
$\delta_{km}\delta_{ij}\cdot(-\frac{P}{p^+})\delta_{jm'}\delta_{ik'}$ and its mirror		$-2\frac{P}{p^+}\delta_{km}\delta_{k'm'}$
$\delta_{km}\delta_{ij}\cdot(-\frac{P}{k^+})\delta_{im'}\delta_{k'j}$ and its mirror		$-2\frac{P}{k^+}\delta_{km}\delta_{k'm'}$
$(\frac{P}{p^+})^2\delta_{jm}\delta_{ik}\delta_{jm'}\delta_{ik'}$		$(\frac{P}{p^+})^2\delta_{mm'}\delta_{kk'}$
$(\frac{P}{k^+})^2\delta_{im}\delta_{kj}\delta_{im'}\delta_{k'j}$		$(\frac{P}{k^+})^2\delta_{mm'}\delta_{kk'}$
$\frac{P^2}{p^+k^+}\delta_{jm}\delta_{ik}\delta_{im'}\delta_{k'j}$ and its mirror		$2\frac{P^2}{p^+k^+}\delta_{mk'}\delta_{km'}$

Only the first product contains a trace. Multiplying by $\frac{p^+k^+}{P^4}$, with $p^+ + k^+ = P$:

$$\frac{p^+k^+}{P^4}\sum_{ij} V_{ij;km}V_{ij;k'm'} = \Big[\frac{d_\perp p^+k^+}{P^4} - \frac{2}{P^2}\Big]\delta_{km}\delta_{k'm'} + \Big[\frac{k^+}{p^+P^2} + \frac{p^+}{k^+P^2}\Big]\delta_{mm'}\delta_{kk'} + \frac{2}{P^2}\,\delta_{mk'}\delta_{km'}. \tag{2.3}$$

This was checked numerically for $d_\perp = 2, 3, 4, 5$ (max. difference 7×10^{-15}).

2.3 Step 3: the w and w' integrations in $d_\perp$ dimensions

In d transverse dimensions:

- the splitting coefficient $\mathbb{C}_1$ carries $\delta^{(d)}$;
- its kernel is the d-dimensional Fourier transform of $\mathbf{q}^m/\mathbf{q}^2$:

$$\mathcal{K}_d^m(r) \equiv -2\pi i \int \frac{d^dq}{(2\pi)^d} e^{iq\cdot r}\frac{q^m}{q^2} = \pi^{1-\frac{d}{2}}\,\Gamma\Big(\frac{d}{2}\Big)\frac{r^m}{|r|^d}, \qquad \mathcal{K}_2(r) = \mathcal{K}(r). \tag{2.4}$$

The argument of the first δ is $\frac{k^+}{P}\big(w_0 - w\big)$ with $w_0 = \frac{P}{k^+}y - \frac{p^+}{k^+}z$. Hence

- $\int d^dw\,\delta^{(d)}[\dots] = (\frac{P}{k^+})^d$;
- $w - z = \frac{P}{k^+}(y - z)$, and $\mathcal{K}_d$ has degree $1 - d$, so $\mathcal{K}_d(w - z) = (\frac{P}{k^+})^{1-d}\mathcal{K}_d(y - z)$.

Per side the factor is $(\frac{P}{k^+})^d(\frac{P}{k^+})^{1-d} = \frac{P}{k^+}$, so both sides give $(\frac{P}{k^+})^2$ *for any* d. In particular there is no $(\frac{P}{k^+})^{-\epsilon}$, which would change the finite part. The phase is $e^{-ik\cdot(w'-w)} = e^{-ik\cdot\Delta\,P/k^+} = e^{-ik\cdot\Delta/\xi}$.

Eq. (2.3) times $(\frac{P}{k^+})^2$, with $\frac{1}{k^+}$ taken out, is your bracket of Eq. (1.13):

$$\frac{1}{k^+}\Big\{\underbrace{\Big[\frac{d_\perp p^+}{P^2} - \frac{2}{k^+}\Big]}_{c_1}\delta_{km}\delta_{k'm'} + \underbrace{\Big[\frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big]}_{c_2}\delta_{mm'}\delta_{kk'} + \underbrace{\frac{2}{k^+}}_{c_3}\delta_{mk'}\delta_{km'}\Big\}. \tag{2.5}$$

2.4 Step 4: the Wilson lines

As in your Eq. (1.14), and checked before (`Groups_II_III_check.pdf`), the colour factor reduces to $N_c\mathcal{O}$. So

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee -k^+}\frac{dp^+}{2\pi}\int_{x,x',y,y'} e^{-ik\cdot\Delta/\xi}\,X^kX'^{k'}\big[c_1\delta_{km}\delta_{k'm'} + c_2\delta_{mm'}\delta_{kk'} + c_3\delta_{mk'}\delta_{km'}\big]J^{mm'}\,\mathcal{O}, \tag{2.6}$$

with the z integral

$$J^{mm'} \equiv \int d^dz\,\mathcal{K}_d^m(y - z)\,\mathcal{K}_d^{m'}(y' - z) \quad \times\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\epsilon/2}. \tag{2.7}$$

2.5 Step 5: the tensor z integral in d dimensions

With Eq. (2.4), $\int d^dz\,e^{-i(q+q')\cdot z} = (2\pi)^d\delta^{(d)}(q + q')$ gives

$$J^{mm'} = (-2\pi i)^2\int\frac{d^dq}{(2\pi)^d}\frac{q^m(-q^{m'})}{q^4}\,e^{iq\cdot(y-y')} = 4\pi^2\int\frac{d^dq}{(2\pi)^d}\frac{q^mq^{m'}}{q^4}\,e^{-iq\cdot\Delta}. \tag{2.8}$$

Use

- $\int\frac{d^dq}{(2\pi)^d}\frac{e^{iq\cdot r}}{(q^2)^2} = \frac{\Gamma(\frac{d}{2}-2)}{16\pi^{d/2}}(r^2)^{2-\frac{d}{2}}$;
- $q^mq^{m'} \to -\partial^m\partial^{m'}$;
- $\partial^m\partial^{m'}(r^2)^a = 2a\,\delta^{mm'}(r^2)^{a-1} + 4a(a-1)\,r^mr^{m'}(r^2)^{a-2}$;
- and, with $a = 2 - \frac{d}{2}$, $a\Gamma(\frac{d}{2} - 2) = -\Gamma(\frac{d}{2} - 1)$ and $a(a-1)\Gamma(\frac{d}{2} - 2) = \Gamma(\frac{d}{2})$.

Then

$$J^{mm'} = \frac{\pi^{2-\frac{d}{2}}}{2}\,(\Delta^2)^{1-\frac{d}{2}}\Big(\frac{\mu^2e^{\gamma_E}}{4\pi}\Big)^{\epsilon/2}\Big[\Gamma\Big(\frac{d}{2} - 1\Big)\,\delta^{mm'} - 2\,\Gamma\Big(\frac{d}{2}\Big)\,\hat\Delta^m\hat\Delta^{m'}\Big] \equiv A\,\delta^{mm'} + B\,\hat\Delta^m\hat\Delta^{m'}. \tag{2.9}$$

The integral converges for $2 < d < 4$ ($\epsilon < 0$): the pole is infrared. With $d = 2 - \epsilon$:

$$A = \frac{\pi}{2}\Big(\frac{\Delta^2\mu^2e^{\gamma_E}}{4}\Big)^{\epsilon/2}\Gamma\Big(-\frac{\epsilon}{2}\Big) = -\frac{\pi}{2}E + O(\epsilon), \tag{2.10}$$

$$B = -\pi\Big(\frac{\Delta^2\mu^2 e^{\gamma_E}}{4}\Big)^{\epsilon/2}\Gamma\Big(1-\frac{\epsilon}{2}\Big) = -\pi + O(\epsilon). \tag{2.11}$$

The trace is $J^{mm} = d_\perp A + B$; with $d_\perp = d$ it equals $\pi^{2-\frac{d}{2}}\Gamma(\frac{d}{2}-1)(\Delta^2)^{1-\frac{d}{2}}(\dots)^{\epsilon/2}$, the scalar integral of Group I (Row II, Row V part 2b).

Numerical checks of Eq. (2.9) by quadrature:

- $d = 2$: the traceless part $J^{11} - J^{22}$ for $\Delta \parallel \hat{e}_1$ is -3.141592653590 (closed form $B = -\pi$);
- $d = 3$: the trace is 2.416609733531 (closed form $\pi/|\Delta| = 2.416609733531$) and $\hat{\Delta}\cdot J\cdot\hat{\Delta} = 2.8e-17$ (closed form 0).

2.6 Step 6: the contraction ($\delta^{mm} = d_\perp$)

- $c_1\,\delta_{km}\delta_{k'm'}$: $X^k X'^{k'} J^{kk'} = A\,X\cdot X' + B\,(X\cdot\hat{\Delta})(X'\cdot\hat{\Delta})$.
- $c_3\,\delta_{mk'}\delta_{km'}$: $X^k X'^{k'} J^{k'k}$. This is the same, because J is symmetric.
- $c_2\,\delta_{mm'}\delta_{kk'}$: $X\cdot X'\,J^{mm} = X\cdot X'\,(d_\perp A + B)$.

Since $c_1 + c_3 = \frac{d_\perp p^+}{P^2}$, the $\frac{2}{k^+}$ terms of your bracket cancel for any $d_\perp$:

$$\Sigma \equiv \frac{d_\perp p^+}{P^2}\Big[A\,X\cdot X' + B\,(X\cdot\hat{\Delta})(X'\cdot\hat{\Delta})\Big] + \Big[\frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big]\big(d_\perp A + B\big)\,X\cdot X'. \tag{2.12}$$

This is exact in d. Inserting A and B:

$$\Sigma = -\frac{\pi}{2}\Big\{X\cdot X'\Big[d_\perp\Big(\frac{p^+}{P^2} + \frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big)E + 2\Big(\frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big)\Big] + 2\,d_\perp\frac{p^+}{P^2}(X\cdot\hat{\Delta})(X'\cdot\hat{\Delta})\Big\} + O(\epsilon). \tag{2.13}$$

The "+2" is B in the trace J^{mm}. The $O(\epsilon)$ terms of A and B multiply $d_\perp$, which is finite, so they drop.

2.7 Step 7: $p^+ \to \xi$

We have $p^+ = k^+\frac{1-\xi}{\xi}$, $dp^+ = -k^+\frac{d\xi}{\xi^2}$, and the limits $\Lambda < p^+ < \vee - k^+$ become $\frac{k^+}{\vee} < \xi < \frac{k^+}{k^+ + \Lambda}$. Then

$$k^+\Big(\frac{p^+}{P^2} + \frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big) = \hat{\Phi}(\xi), \qquad k^+\Big(\frac{1}{p^+} + \frac{p^+}{k^{+2}}\Big) = \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}, \qquad k^+\frac{p^+}{P^2} = \xi(1-\xi). \tag{2.14}$$

Collecting the prefactors, $\frac{g^4 N_c}{4\pi^4}\cdot\frac{1}{2\pi}\cdot\big(-\frac{\pi}{2}\big) = -\frac{g^4 N_c}{16\pi^4}$. With $E = -\bar{E}$ this gives Eq. (1.5).

2.8 Step 8: the endpoint

- The only singular coefficient at $\xi \to 1$ is $\frac{\xi}{1-\xi} = \frac{1}{1-\xi} - 1$, in $\hat{\Phi}$ and in the $d_\perp$-independent term. Together it multiplies $(d_\perp E + 2)\,X\cdot X'$.
- Exactly, $\int_{\xi_0}^{\xi_1} d\xi\frac{h(\xi)}{1-\xi} = \int_{\xi_0}^{\xi_1} d\xi\frac{h(\xi)-h(1)}{1-\xi} + h(1)\log\frac{1-\xi_0}{1-\xi_1}$.
- The limits give

$$\log\frac{1-\xi_0}{1-\xi_1} = \log\frac{(\vee - k^+)(k^+ + \Lambda)}{\vee\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log\Big(1 - \frac{k^+}{\vee}\Big) + \log\Big(1 + \frac{\Lambda}{k^+}\Big). \tag{2.15}$$

This gives Eq. (1.6). It was checked numerically for arbitrary $d_\perp$ and $\bar{E}$ (3×10^{-16}).

2.9 Step 9: the pole as a convolution with LO

- $\frac{d^3 N_{\rm LO}}{d^3 k}\big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\big)$ has the prefactor $\frac{g^2\xi}{2\pi^2 k^+}$ and the phase $e^{-ik\cdot\Delta/\xi}$.
- So $\int e^{-ik\cdot\Delta/\xi} X\cdot X'\,\mathcal{O} = (2\pi)^3\frac{2\pi^2 k^+}{g^2\xi}\frac{d^3 N_{\rm LO}}{d^3 k}\big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\big)$.
- The $\frac{1}{\epsilon}$ part of Eq. (1.5) is then $\frac{g^2 N_c}{8\pi^2}\frac{1}{\epsilon}\int\frac{d\xi}{\xi^3}d_\perp\hat{\Phi}\,\frac{d^3 N_{\rm LO}}{d^3 k}$, which is Eq. (1.8) with $\frac{g^2 N_c}{8\pi^2} = \frac{\alpha_s N_c}{2\pi}$.
- The measure is $d\xi/\xi^3$.

3 Checks

Script `check_d.py`, results in `check_d.json`.

check	result
polarization sum (2.3), $d_\perp = 2, \ldots, 5$	max. difference 7×10^{-15}
z integral (2.9), $d = 2$ traceless part	-3.141592653590 vs. $-\pi$
z integral (2.9), $d = 3$ trace and $\hat{\Delta} J \hat{\Delta}$	rel. 2×10^{-14}
contraction (2.12) with $J = A\delta + B\hat{\Delta}\hat{\Delta}$, $J^{mm} = d_\perp A + B$ (`sympy`)	difference 0
$d_\perp = 2 - \epsilon$: reduces to the earlier result (`sympy`)	difference 0
ξ forms of the coefficients (`sympy`)	differences 0, 0, 0
endpoint decomposition (1.6), exact limits	rel. 3×10^{-16}